CH 4 - Inner Product Spaces

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We defined abstract vector spaces to model $\mathbb{R}^n$.

So far, the focus has been on algebra. Geometry is rarely discussed.

In $\mathbb{R}^n$, we can talk about lengths, distance, and angles via DOT PRODUCT

Recall that in $\mathbb{R}^n$, $\vec{x} \cdot \vec{y} = \begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ \dots \\ y_n \end{bmatrix} = x_1 y_1 + \dots + x_n y_n$

Observations:

1. $\vec{x} \cdot \vec{x} = x_1^2 + \dots + x_n^2 = \|\vec{x}\|^2$
2. $\text{dist}(x, y) = \|\vec{y} - \vec{x}\| = \sqrt{(\vec{x} - \vec{y}) \cdot (\vec{x} - \vec{y})}$
3. $\vec{x} \cdot \vec{y} = \|x\|\|y\| \cos\theta \implies \theta = \arccos \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\|\|\vec{y}\|}$

We can view dot product as $f : V \times V \to \mathbb{R}$, $f(x, y) = x \cdot y \in \mathbb{R}$ that satisfies

1. $f(x, y) = f(y, x)$, symmetry
2. $f(cx + z, y) = cf(x, y) + f(z, y)$, linearity in the first argument(second argument is same by symmetry)
3. $f(x, x) \geq 0 \ \forall x$, $f(x, x) = 0 \iff x = 0$, positive-definite
4. $\frac{f(x,y)}{\|x\|\|y\|} \in [-1, 1]$, can be obtained from the above 3 properties.

📘 **Definition — Inner Product**

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